

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Project Name: ComplaintEase – Advanced Complaint Management Platform

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Email and Password
FR-2	User Authentication	JWT Login and Session Management
FR-3	Complaint Submission	Create Complaint with Category, Priority and Description
FR-4	File Management	Upload Images and Documents via Cloudinary
FR-5	Complaint Tracking	Track Complaint Status Timeline
FR-6	Comment System	Add Comments and Discussion Updates
FR-7	Notifications	Real-Time Notifications using Socket.io
FR-8	Complaint Rating	Rate Resolved Complaints
FR-9	User Management	Manage User Accounts and Roles
FR-10	Department Management	Assign Complaints to Departments
FR-11	Analytics Dashboard	View Complaint Trends and KPIs
FR-12	Report Generation	Generate PDF and CSV Reports
FR-13	Audit Logging	Track Administrative Activities

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
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NFR-1	Usability	Simple and user-friendly interface for all users.
NFR-2	Security	JWT authentication, role-based access, data protection.
NFR-3	Reliability	System should operate consistently without failures.
NFR-4	Performance	Response time should remain below acceptable limits.
NFR-5	Availability	System available 24x7 with minimal downtime.
NFR-6	Scalability	Support increasing users, complaints, and departments.
NFR-7	Maintainability	Easy updates and modular code structure.
NFR-8	Portability	Accessible across desktop, tablet, and mobile devices.